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Enhancing DDoS Resilience in Government Voter Systems: Extending Google Project Shield with Real-Time Anomaly-Aware L4–L7 Adaptive Filtering

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Abstract

In recent years, election systems have become one of the main targets for cyberattacks, especially Distributed Denial of Service (DDoS) attacks. Google's Project Shield is a free service that helps protect public-interest websites, including election-related ones, from attacks at the application layer (Layer 7). However, it does not protect traffic at the transport layer (Layer 4), which leaves important parts like voter registration systems and public APIs exposed to large-scale traffic floods. This research introduces a cloud-based, AI-powered adaptive filtering system that works together with Project Shield to add protection at both L4 and L7. The system uses behavior-based detection to learn normal traffic patterns during quiet periods and automatically applies filtering rules when suspicious traffic is detected. This project will evaluate the system's accuracy, response time, and how well it avoids blocking real users using simulated attacks that reflect actual election-day conditions. The goal is to create a more reliable and efficient security solution for public election websites one that doesn't require extra hardware and can be used even by governments with limited resources.

Key Terms

Election Security, DDoS Mitigation, L4–L7 Filtering, Adaptive AI, Project Shield, API Protection, Zero Downtime

1. Introduction

Election platforms are vital tools for supporting democratic participation and managing civic duties. As governments increasingly rely on digital systems for elections, these platforms face growing exposure to a range of cybersecurity threats. One of the most serious risks is Distributed Denial of Service (DDoS) attacks, which aim to flood systems with excessive traffic, rendering them unavailable to legitimate users. These attacks are often planned around crucial moments like voter registration deadlines or election days when many people try to access the systems at once. This makes it easier for attackers to cause disruptions that prevent voters from registering or verifying their identities. Services such as public voter portals and registration APIs are often the most targeted, as they are public-facing and can be overwhelmed quickly. For governments, maintaining availability, transparency, and security during these critical windows is becoming more difficult due to the increasing scale and complexity of attacks.

Google's Project Shield is one free solution that offers Layer 7 (L7) DDoS protection for websites that serve the public interest, especially those related to elections. It can effectively block common web-layer attacks like HTTP GET floods by filtering application-level traffic. However, a major limitation is that it does not offer native protection for Layer 4 (L4) traffic, such as TCP or UDP requests. These lower-layer

attacks are designed to flood bandwidth or exhaust the target's transport layer resources before they reach application-level defenses (Doriguzzi-Corin & Siracusa, 2022). This oversight leaves a critical gap in protection, particularly for endpoints like voter APIs and HTTPS connections, which are essential during election periods. Election bodies in developing or underfunded regions often cannot afford the added cost of L4 hardware appliances, making this vulnerability even more serious. Without L4 protection, these systems are exposed to traffic floods that can take down services completely.

Most government election systems still depend on outdated and static security configurations, including simple rate-limiting or content delivery network (CDN) rerouting (Radware, 2025). These methods often rely on fixed thresholds that may not adjust properly during high but legitimate traffic loads, such as during voter verification close to deadlines. If those thresholds are too low, valid users may be blocked. If they're too high, attacks might pass through undetected. These older techniques also cannot learn or evolve in response to changing attack behavior. As a result, attackers who test and adjust their strategies can bypass static filters over time. Meanwhile, real users experience slowdowns, blocked access, or complete denial of service. This makes legacy solutions insufficient for today's fast-moving and adaptive cyber threats targeting digital election infrastructure.

Although cyberattacks against election systems are growing in number and complexity, innovation in cloud-based and adaptive defense systems has been slow, particularly in government environments. Many enterprise-level cloud services now offer advanced DDoS protections that include L4 and L7 filtering, but these come

with subscription costs or require private infrastructure, making them less accessible to public election commissions. In contrast, many public systems still rely on basic free tools that do not provide complete stack protection (Yigit et al., 2023). Election systems are now more dynamic than ever, using APIs, supporting real-time updates, and being accessed by various devices. Without intelligent, scalable defense mechanisms, election bodies risk large-scale disruptions that can prevent citizens from voting or reduce confidence in the fairness of digital election platforms.

This research proposes an advanced yet accessible solution that improves upon Project Shield by adding an AI-driven adaptive filtering system capable of analyzing both L4 and L7 traffic in real time. The proposed model includes mechanisms that learn normal traffic behaviors during quiet times and then apply mitigation rules only when deviations are detected. These mitigation actions are deployed directly to Google's edge infrastructure, which helps block malicious traffic before it even reaches the voter systems (Ahmed et al., 2023). This approach removes the need for expensive hardware appliances and manual updates, allowing even smaller election commissions to defend themselves using automated, cloud-native systems. With this enhancement, voter-facing services can remain online and responsive even during peak periods or coordinated attack attempts. By offering a scalable and cost-effective design, the system aims to significantly strengthen the security and reliability of digital election platforms.

2. Problem Statement

Government election systems and digital voting platforms are becoming bigger targets for cyberattacks, especially during important

voting times. Among the most disruptive methods are Distributed Denial of Service (DDoS) attacks, which aim to disable or degrade the availability of public-facing services like voter registration portals, election results dashboards, and authentication APIs. Google's Project Shield offers substantial Layer 7 (L7) protection, efficiently mitigating HTTP-based floods. However, its inability to filter Layer 4 (L4) traffic such as TCP SYN floods and UDP floods results in a major vulnerability. These attacks can overwhelm server sockets and saturate bandwidth even before requests are processed by application-layer defenses (Nokia, 2024). Consequently, government systems remain vulnerable to a class of attacks that target the network and transport layers directly, leading to outages and damaged public trust.

To address this vulnerability, this research investigates and proposes a next-generation defense mechanism that combines AI-based anomaly detection with cloud-native traffic filtering to enhance Project Shield's capabilities (Nokia & NL-ix, 2024). The proposed solution leverages machine learning algorithms to establish baseline traffic behaviors during non-peak periods, such as IP frequency distribution, protocol usage patterns, and connection duration histograms. When deviations from this baseline are detected during live traffic analysis, adaptive rule sets are automatically generated and deployed to Google's edge network. This enables real-time mitigation of volumetric floods without the need for customer-side appliances or manual rule configuration. This system not only closes the current L4 defense gap but also introduces a more intelligent, scalable model for protecting government election services during times of peak vulnerability.

3. Research Aim

This research aims to investigate DDoS vulnerabilities affecting digital voter systems in government infrastructure and propose a scalable, AI-driven L4–L7 filtering enhancement to Google Project Shield that ensures zero-downtime API protection.

4. Research Objectives

RO1: To identify and analyze DDoS vulnerabilities affecting government election platforms, especially at the transport layer.

RO2: To evaluate the functional and architectural limitations of Google Project Shield and comparable cloud-based protections.

RO3: To design and simulate an adaptive filtering system using anomaly detection for L4 and L7 protection at the cloud edge.

5. Research Questions

RQ1: What are the recurring DDoS threats targeting voter-facing APIs and endpoints within public election systems?

RQ2: What are the limitations of current mitigation platforms like Google Project Shield in protecting transport-layer traffic?

RQ3: How can adaptive anomaly detection be effectively applied to enhance L4–L7 DDoS protection without deploying additional hardware?

6. Research Significance

- **To significantly improve the digital security posture of public election systems** by proposing a solution that bridges the critical L4 protection gap present in most freely available cloud services. By incorporating adaptive traffic monitoring and dynamic response mechanisms, the proposed system helps to ensure uninterrupted

access to voter services during peak periods, reinforcing public confidence in the democratic process (IEEE TDSC, 2023). The scalability and ease of integration into existing platforms further ensure that even smaller governmental bodies can implement advanced protection without a steep technical or financial burden.

- **To introduce an efficient and cost-effective cybersecurity solution for election commissions** that may lack technical infrastructure or resources to deploy enterprise-grade DDoS defenses. By eliminating dependency on physical appliances and offering automated detection and mitigation workflows, the solution empowers government IT teams to handle evolving threats with minimal intervention. This democratization of

high-level cyber defense contributes to greater equity in electoral security across different regions and institutions. (Hamarshe et al., 2023)

- **To support broader digital governance objectives** by ensuring that election technologies align with current and emerging cybersecurity standards. The use of AI for adaptive defense, combined with real-time deployment at the cloud edge, helps future-proof public digital infrastructure against new forms of DDoS attacks. It also provides a foundation for scalable enhancements, such as federated anomaly sharing and transparent audit mechanisms, aligning with ongoing global efforts to secure democratic institutions in the digital era.

7. Overview of the Proposed System

The proposed system introduces an AI-powered, cloud-native solution that enhances Google Project Shield's ability to protect government voter systems from modern Distributed Denial of Service (DDoS) attacks (Springer Info Sys Frontiers, 2024). Unlike conventional static defenses, this system actively learns from network behavior, detects anomalies in real-time, and deploys filtering rules dynamically at the edge, ensuring zero-downtime availability for APIs and voter-facing platforms.

The core of this system is built on three main modules: the **Baseline Profiling Engine**, the **Anomaly Detection and Filtering Module**, and the **Edge Deployment Interface**. Each of these components contributes to the continuous monitoring, learning, and response processes that help mitigate

sophisticated DDoS attacks while maintaining service availability for legitimate users.

During normal or non-election periods, the **Baseline Profiling Engine** analyzes incoming network traffic and identifies consistent patterns using unsupervised machine learning. It considers data features such as average packet size, IP request frequency, connection time, and protocol types (e.g., TCP, UDP, HTTPS). These patterns form a behavioral baseline that represents "normal" usage of the voter system APIs and portals. This phase enables the system to later distinguish malicious anomalies from natural high-traffic events like flash crowds.

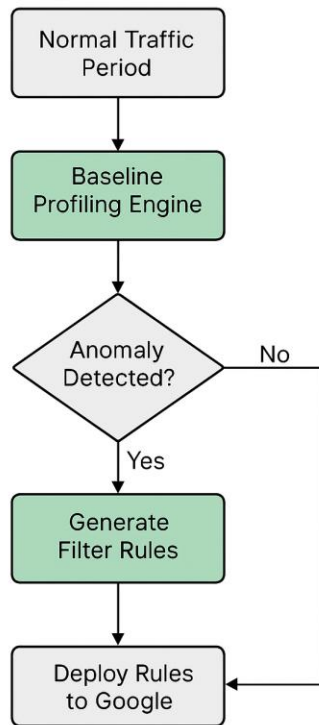


Figure 1: Flowchart for Adaptive Anomaly-Based DDoS Detection and Mitigation

When an election event begins or when user traffic surges, the system activates its Anomaly Detection Module. This module compares live traffic to the stored baseline model. If a particular IP address or group of addresses exceeds expected behavior such as sending repeated requests within milliseconds, initiating traffic spikes using uncommon protocols, or producing payloads that don't match known patterns the module flags them as suspicious. It then initiates mitigation using the Filtering Rule Generator, which generates temporary but intelligent rules (Fenil & Kumar, 2022). These rules might include dropping UDP packets, rate-limiting specific IP addresses, blacklisting subnets, or even geofencing traffic for short periods.

The **Edge Deployment Interface** enables these mitigation rules to be sent instantly to Google's globally distributed edge infrastructure, rather than filtering traffic locally on election servers. This design ensures that malicious traffic is blocked close to its origin, far before it reaches the voter infrastructure. By working in tandem with Project Shield's existing L7 defenses, this added L4 intelligence allows for a full-stack protection model. Moreover, because no customer hardware is required, even low-budget election commissions can deploy this system with minimal effort.

Once the rules are active, the system continues monitoring the network. If traffic stabilizes or if the system recognizes that it misclassified legitimate traffic, the rules expire automatically or are revised. This ensures that the system remains flexible, avoids long-term blocking, and continues learning to adapt to new patterns. The system is designed to continuously evolve its baseline through feedback loops improving detection accuracy over time.

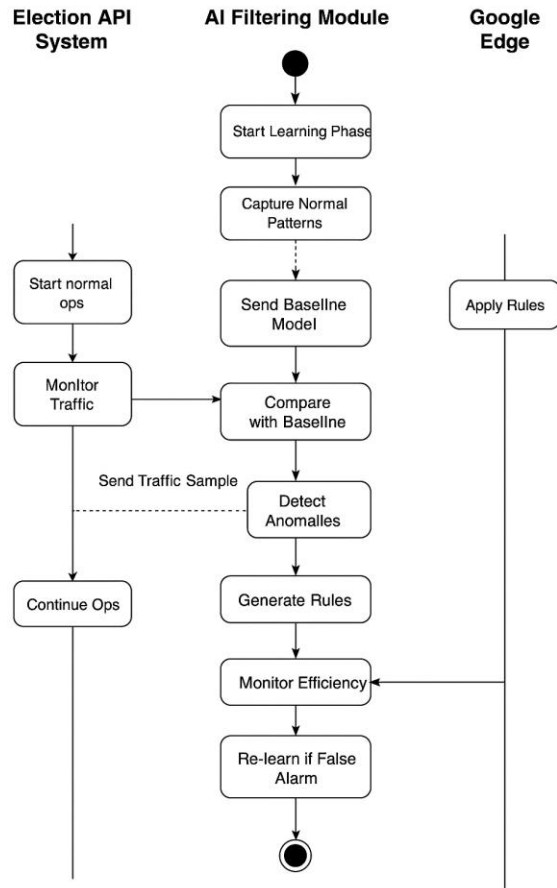


Figure 2: Activity Diagram for Anomaly-Based Filtering Interaction in Government Voter Systems

To better understand how each part of the system works together, it helps to look at it using a class-based structure. The system is built using modular classes that follow object-oriented programming principles. The `TrafficSample` class is responsible for capturing real-time traffic data, including details like timestamp, source IP, destination IP, and the number of packets. These samples are then collected by the `SampleAggregator`, which passes them to the `AIFilteringModule`. This module checks the incoming data against a reference model called `BaselineModel`. If it finds anything unusual, it triggers a rule using the `IRule` interface. This interface contains the instructions needed to handle the threat, such as slowing

down traffic (throttling), blocking it (dropping), or redirecting it.

This class-based design makes the system easier to manage and expand. For instance, if another government or organization wants to add its own filtering rules, it can do so by extending the `IRule` interface. This makes the system flexible enough to adapt to different policies or use cases. It also helps integrate the system smoothly with existing security setups, especially for governments already using platforms like Google Cloud or public-facing API services.

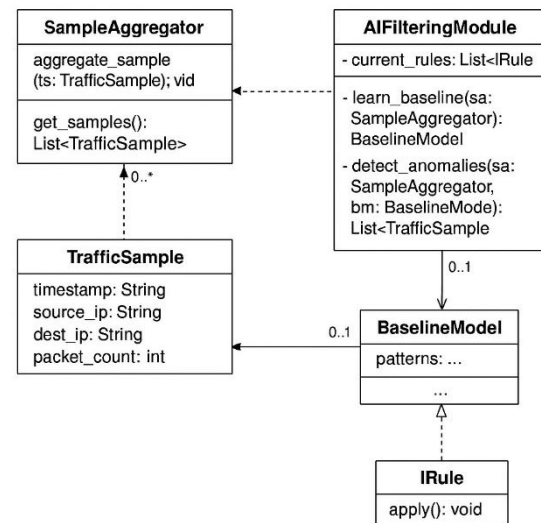


Figure 3: UML Class Diagram for Anomaly-Based Traffic Analysis and Rule Generation

One of the key strengths of this proposed system is that it follows the Observe, Orient, Decide, and Act (OODA) cycle. It keeps an eye on incoming traffic all the time, checks it against what's considered normal behavior, takes action by applying the right filtering rules, and then learns from the outcome to make future decisions in the future. This loop allows the system to keep up with changing attack patterns and stay effective even as

cyber threats become more advanced. It also reduces the need for constant human monitoring, since most of the traffic detection and response happens automatically.

Another big advantage of the system is its ability to scale. Many election commissions in smaller countries or regions don't have access to expensive protection tools like scrubbing centers or round-the-clock cybersecurity teams. This system solves that issue by offering a high level of protection through the cloud. It makes strong security more accessible, giving every election body a fair chance to protect their digital platforms, regardless of their budget or available technology.

The system is also designed with privacy in mind. It doesn't collect or store any personal information from voters or election processes. Instead, it only observes traffic patterns and behavior. This helps the system stay in line with privacy laws and public expectations, building trust in the use of technology during elections. In the future, this design could be extended to include federated anomaly learning, where different election organizations can anonymously share behavioral patterns. This would allow them to improve their defenses together without ever sharing private or sensitive data.

To sum up, this AI-based adaptive filtering system offers a major improvement in protecting government election platforms, especially by filling the security gap left by existing cloud tools that don't include Layer 4 protection. With features like real-time learning, fast anomaly detection, and edge-level deployment, the system provides strong, full-layer defense without needing extra equipment. The use of clear visual tools like flowcharts, activity diagrams, and class models also helps explain the system and guide its development for real-world use.

8. Conclusion

As more democratic processes move online, keeping election systems available and secure at all times has become a top priority for governments around the world. This research has highlighted a serious weakness in current DDoS protection strategies specifically, the lack of Layer 4 (L4) defense in widely used services like Google Project Shield. When attackers focus on APIs, registration systems, or lower-level network connections, they can cause major disruptions, even if Layer 7 (L7) protections are in place. These attacks can block voter access or delay important services, which puts pressure on the fairness and reliability of digital elections.

To solve this issue, this project suggests an AI-powered adaptive filtering system that works alongside Project Shield. The system learns how normal traffic behaves and watches for unusual patterns. When it sees something suspicious, it immediately applies custom filtering rules at the edge of the network. This makes the solution fast, cost-effective, and able to scale for different-sized election agencies. By automating the entire process from learning to reacting to the system also reduces the need for constant human monitoring, which is important during busy and high-risk times like elections.

One of the key strengths of this system is its modular, cloud-based design. This means it can be updated easily and used in different regions or setups without much effort. It uses machine learning and a feedback loop to improve itself over time, so it stays effective even as new attack methods appear. Because it doesn't rely on extra hardware, it is also more affordable and accessible to smaller government organizations. This helps build

public trust in digital voting systems by keeping them reliable and secure.

Looking ahead, there are many ways this system could become even more powerful as technology advances. For example, federated learning could let different election authorities share anonymized data to improve detection models without giving up privacy. Blockchain technology could be added to record all filtering actions in a secure and unchangeable way, making it easier to track and audit decisions. Predictive analytics could be used to spot potential attacks before they happen, and zero-trust frameworks could help protect both public systems and internal networks. As cybersecurity standards continue to develop, this solution could become the core of a smarter, more adaptive way to protect elections in the digital age.

(2932 words)

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